Understanding Line Search Optimization

Binary Search & Golden-Section Search

Study Guide for Assignment Part (e)

The Goal of Part (e)

In previous parts, you calculated the **descent direction** ($d = -\nabla J$) using the gradient. You know *which way* to move to reduce the loss, but you don't know *how far* to move.
Part (e) asks you to find the optimal step size η (eta). We define a 1-dimensional function representing the loss along that line:

$$\phi(\eta) = J(w_{current} + \eta \cdot d)$$

Your task is to minimize $\phi(\eta)$ over the interval $[0, 1]$.

1 Method 1: Binary Search (Dichotomous Search)

1.1 The Intuition: The Blindfolded Hiker

Imagine you are standing in a valley, blindfolded. You want to find the bottom. You stand at the center of the search area (midpoint m).

- You tap the ground slightly to your right ($m + \epsilon$).
- If the ground goes **UP**, the bottom must be behind you (Left).
- If the ground goes **DOWN**, the bottom must be ahead of you (Right).

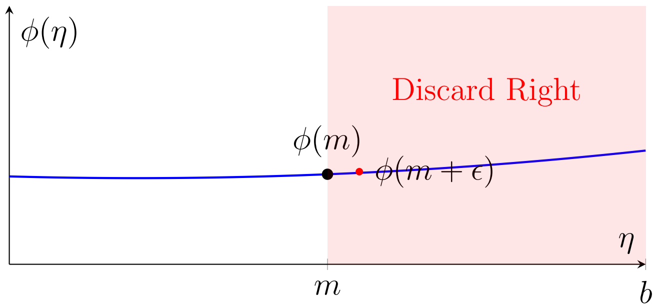
1.2 The Algorithm

Given an interval $[a, b]$:

1. Calculate the midpoint $m = \frac{a+b}{2}$.
2. Evaluate the function at m and at $m + \epsilon$ (where ϵ is a small number like 0.001).
3. **Update Rule:**
   - **Case A:** If $\phi(m) < \phi(m + \epsilon)$, the slope is increasing. The minimum is to the **Left**.
     New Interval: $[a, m]$.

- **Case B:** If $\phi(m) > \phi(m+\epsilon)$, the slope is decreasing. The minimum is to the **Right**.
  New Interval: $[m, b]$.

Visualizing Binary Search (Left Case)



Practice Example: Minimizing $\phi(\eta) = (\eta - 0.3)^2 + 5$ on $[0, 1]$

Iteration 1:

- Midpoint $m = 0.5$. Test point 0.501.
- $\phi(0.5) = (0.2)^2 + 5 = 5.04$.
- $\phi(0.501) = (0.201)^2 + 5 = 5.0404$.
- **Comparison:** $5.04 < 5.0404$. The function is increasing.
- **Result:** Go Left. New Interval is $[0, 0.5]$.

2 Method 2: Golden-Section Search (1/4 - 3/4 Variant)

2.1 The Intuition

Calculating derivatives or slopes can be noisy. Instead, we pick two interior points and simply compare their heights. We keep the region surrounding the lower point.

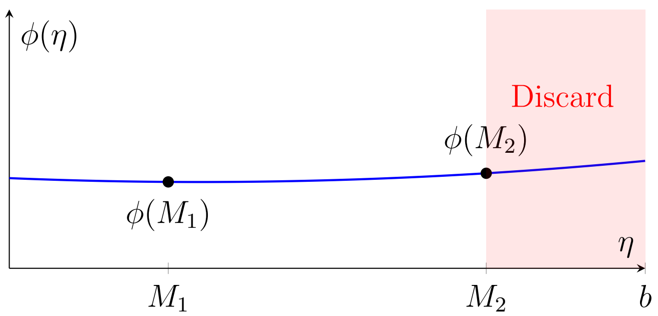
2.2 The Algorithm

Note: Your assignment uses the simplified fractions $1/4$ *and* $3/4$ *rather than the standard golden ratio.*

Given an interval $[a, b]$:

1. Calculate two test points:
$$M_1 = a + 0.25(b - a) \quad \text{and} \quad M_2 = a + 0.75(b - a)$$
2. Evaluate $\phi(M_1)$ and $\phi(M_2)$.
3. **Update Rule:**
   - **If** $\phi(M_1) < \phi(M_2)$**:** The minimum is closer to M_1 (Left). Eliminate the far right segment.
     New Interval: $[a, M_2]$.
   - **If** $\phi(M_1) > \phi(M_2)$**:** The minimum is closer to M_2 (Right). Eliminate the far left segment.
     New Interval: $[M_1, b]$.

Visualizing Golden Search (Left Case)



Practice Example: Minimizing $\phi(\eta) = (\eta - 0.3)^2 + 5$ on $[0, 1]$

Iteration 1:

- $M_1 = 0.25, M_2 = 0.75$.
- $\phi(0.25) = 5.0025$.
- $\phi(0.75) = 5.2025$.
- **Comparison:** $\phi(M_1) < \phi(M_2)$. The minimum is near M_1.
- **Result:** Discard the far right $[0.75, 1]$. New Interval: $[0, 0.75]$.

3 Roadmap to Solving Part (e)

Do not just plug in numbers blindly. You must derive the equation first.

Step 1: Construct the Equation $\phi(\eta)$

1. **Identify Inputs:**
   - Current weights $w = [w_1, w_2]^T$ (from Part c).
   - Descent direction $d = [d_1, d_2]^T$ (this is usually $-\nabla J$).
2. **Parametrize the Weights:** Write the new weights as a function of η:
$$w_{new}(\eta) = w + \eta d = \begin{bmatrix} w_1 + \eta d_1 \\ w_2 + \eta d_2 \end{bmatrix}$$
3. **Substitute into Loss Function:** Plug these expressions for w_1 and w_2 into your original $J(w)$ function.
4. **Simplify:** Expand the algebra. You should end up with a simple quadratic equation:
$$\phi(\eta) = A\eta^2 + B\eta + C$$
Use this simplified equation for all your search calculations.

Step 2: Execute Binary Search

- **Start:** Interval $[0, 1]$.
- **Iter 1:** Check slope at $m = 0.5$. Update interval to $[a_1, b_1]$.
- **Iter 2:** Calculate the **new** midpoint of $[a_1, b_1]$. Check slope again.

Step 3: Execute Golden-Section Search

- **Start:** Interval $[0, 1]$.
- **Iter 1:** Check $M_1 = 0.25, M_2 = 0.75$. Update interval to $[a_1, b_1]$.
- **Iter 2 (Important!):** You cannot reuse points easily with the 0.25/0.75 method. You must calculate new test points based on the new interval length:
$$M_1^{new} = a_1 + 0.25(b_1 - a_1)$$
$$M_2^{new} = a_1 + 0.75(b_1 - a_1)$$
Compare and update.